Do Now · retrieval

GPUs, multicore and parallel systems

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why doesn't a smaller assembly instruction count guarantee a faster program?

Q2

What is a core?

Q3

Two independent CPU-bound tasks each take 4 minutes when run on its own core. They start together on two separate identical cores. Calculate the ideal elapsed completion time, ignoring overhead.

Q4

Given int[] a={1,2,3,4}, perform a[i]=a[i]*2 for even indexes 0 and 2 only. Give final array and its sum.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

GPUs and data parallelism — A GPU contains many execution resources suited to repeating similar calculations across graphics pixels or machine-learning tensors.

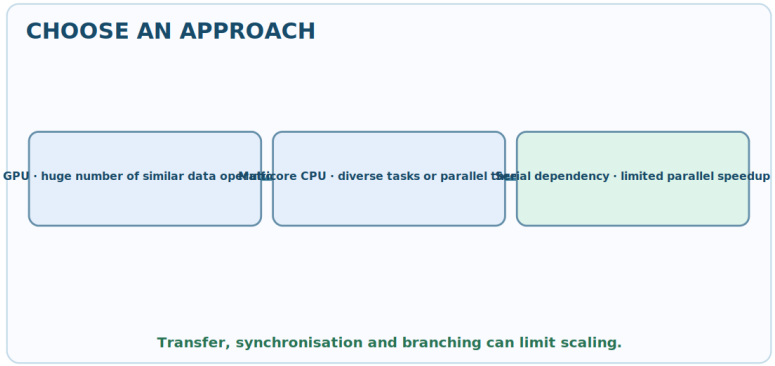
Core idea 2

Multicore CPUs — Multiple CPU cores can run independent threads/processes, enabling greater throughput when tasks can be parallelised.

Core idea 3

Evaluate overhead — Data transfers, branching, synchronisation and sequential tasks can reduce the ideal speedup. A GPU is not automatically best for every problem.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

GPU — Processor designed for highly parallel workloads.

multicore — CPU containing multiple execution cores.

data parallelism — Apply the same operation to many independent data elements.

bandwidth — Volume of data transferred in a given time.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

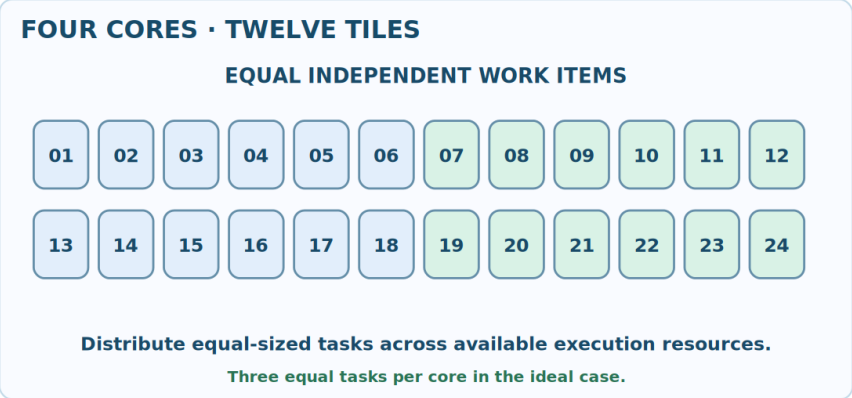
Compare 12 independent image tiles split across four cores under an ideal equal-work model.

Follow the worked method

1. Predict first — Divide 12 independent equal-sized tiles among four available cores.

2. Explain and check — Each core processes three tiles, so ideal completion is around one quarter of serial time only if work and overhead allow.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

The blocks can be divided evenly into five independent groups.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Give two uses of GPUs, including one not based on graphics, and explain a common characteristic of both.

Question 2

A task has 24 independent identical chunks and six CPU cores. Calculate ideal work per core and describe two constraints on real speedup.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A video-editing workflow must first decode frames sequentially, then apply the same filter independently to each frame, before assembling output. Explain which parts suit GPU execution and why.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

GPUs always execute any sequential algorithm faster than a CPU.

Use key vocabulary from memory

1. Explain GPU in your own words.

2. Explain multicore in your own words.

3. Explain data parallelism in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.